

EDUCATION

Stanford University, B.S. Physics, Computational Physics Depth

Expected 2028

- **GPA:** 3.84/4.0
- **Relevant Coursework:** Machine Learning, Computer Vision, Reinforcement Learning, NLP, Computer Organization and Systems, Operating System Principles, Computational Physics.

EXPERIENCE

AI Engineering Intern

June 2026 - Present

Cerence AI

Remote – Burlington, MA

- Built a synthetic data pipeline generating 25K+ labeled examples for a LoRA routing model using generator and judge LLMs with multi-stage quality control (LLM-as-judge with arbitration, rule-based checks, sampled human review) that filtered $\sim 1K$ low-quality samples ($\sim 4\%$).
- Fine-tuned mmBERT (Python/PyTorch) for NLU classification tasks, reaching 95% accuracy across 15 classes.
- Integrated routing model into a multi-agent voice-assistant pipeline via a non-blocking async task with no measurable added latency, implementing interaction-scoped cancellation to safely support user barge-in.
- Cut p95 latency of a LoRA-adapter routing model by 40% (200ms to 120ms) on CPU via ONNX Runtime, dynamic INT8 quantization, and intra-op thread tuning, at a cost of 4% accuracy.
- Investigated the relationship between batching, GPU utilization, KV-cache usage, and latency (primarily time-to-first-token) to determine deployment tradeoffs with a 30B parameter LLM.

Research Intern

June 2025 - September 2025

Molecular Imaging Program at Stanford (MIPS)

Stanford, CA

- Developed a graph neural network with PyTorch for binary classification over PET scan events, implementing encoding layers, Graph-attention layers, classification head, and using residual connections to stabilize training.
- Handled class imbalance through cost-sensitive training and tuned classification threshold using ROC curves achieving 92.9% precision, 92% recall, $F0.5 = 0.927$ on held-out test set.
- Optimized PET scan image reconstruction pipeline with NumPy, cutting processing time by 30% per sample.

PROJECTS

GUI World Models via HTML Generation *Adaptation of Code2World*

2026

- Fine-tuned Qwen3-VL-8B with LoRA to predict the next GUI state (via renderable HTML) given (screenshot, action) pairs. Achieved 40% improvement over base model on LLM-as-judge semantic evaluation.
- Developed data generation + validation pipeline using GPT-4o as generator and SigLIP as validator, as well as a GPT-4o revision loop for rejected entries, to create a dataset of 755 (screenshot+action to next state) pairs.

Custom Heap Allocator

2025

- Built a heap allocator in C to support "malloc", "realloc", and "free" operations for memory management.
- Optimized memory utilization with free-lists and immediate coalescing to reduce fragmentation, achieving 91% utilization on held-out test set covering real-world memory workflows, accepting slight latency increase.

SKILLS

Programming Languages

Python, C++, C, Bash.

ML & Data

PyTorch, Hugging Face Transformers, LoRA, NumPy, Pandas, SciPy

Inference & Serving

vLLM, ONNX Runtime, AIPerf, nvidia-smi

Infrastructure & Tools

Git, Unix, HPC/Slurm, Docker, OpenAI SDK